Pooling and Drift in Delayed Bandits

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Abstract

A system often has to act long before it learns whether the act worked: a recommender sees a click in seconds and a purchase in days. With K actions and a delay of d rounds, the best rate available is $\widetilde{O}(\sqrt{(K+d)T})$ over T rounds (Esposito et al., 2023), so a longer menu is always more expensive to learn from. It need not be. If the outcome depends on the action only through the state it produced, a click or a browse or a cart, then one late outcome informs every action that could have produced the state observed, and the price is set by how many genuinely different states the actions produce rather than by how many actions there are. We measure that by an *effective dimension* v_t between 1 and the number of states, and prove $\widetilde{O}(\sqrt{(d+1)V \log K})$ for a rotating algorithm and $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$ for the single-copy one that is actually used, for any budget fixed in advance; merging similar states lowers the price again, at a bias we make explicit. The idea has a limit. Even when handed the exact losses from d rounds ago, no algorithm escapes $\Omega(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$, where J counts the drifting directions and $\mathcal{E}$ bounds how far the losses move while the learner waits. On generated data the state channel cuts regret by up to 79 per cent against action-level weighting and by 32 to 68 per cent against the tuned minimax-optimal method of Zimmert and Seldin (2020).

1 Introduction

Delayed outcomes are common in sequential decision-making: a system must act now even though the outcome it cares about may arrive much later. In recommendation, an item is shown immediately, an engagement signal such as a click or browsing depth is observed soon after, and conversion or retention may only be known days or weeks later, so decisions are made before their economically relevant consequences appear. The central question is when such a signal can reduce the cost of learning before the final outcome arrives.

The pattern is not special to recommendation: a markdown is set and sell-through is known a season later; staff are rostered and the service outcome lands after the shift; marketing spend is committed before attribution resolves. A fast proxy arrives at once; the outcome the decision is judged on does not. Here an action is an item that could be shown, a state is what the user does next, and the delayed outcome is whether they eventually buy. Different items may induce similar distributions over engagement states, so an outcome observed after one item can carry information about many others. The difficulty scales not with the number of actions K but with how many genuinely different operational regimes those actions induce, so thousands of items funnelling users through a few engagement patterns stay easy to learn from.

Bandits with intermediate observations formalize this: the learner repeatedly picks one action and sees the consequence of that action alone. Esposito et al. (2023) show that the *state-to-loss* map decides the complexity, an unrestricted one giving only the rate available with no intermediate observations; we fix that map and ask how much the *action-to-state* structure helps.

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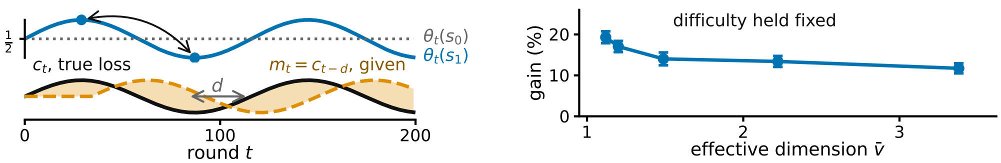

Figure 1: **An illustration of the two limits on intermediate feedback, on generated data. Left,** what each state predicts about the outcome changes while the map from actions to states stays fixed, so the losses from d rounds earlier grow increasingly inaccurate; E_2 of Section 3 tracks it. **Right,** more overlap between actions allows more sharing, at matched difficulty, $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference (Appendix H, study 10).

Intermediate observations face two independent limitations: whether information can be shared *across actions*, and whether it stays useful *across time*.

Across actions: because similar actions reach similar states, we can *pool through the state*, charging a delayed outcome to every action that could have produced the state observed. The resulting statistical cost is governed by an *effective dimension* v_t, which measures how distinguishable the action-induced state distributions are under the learner's current play. When many actions behave similarly through the state channel, v_t can be far smaller than K.

Across time: delay creates a second difficulty even when the learner is handed an exact description of the past. Let $m_t = c_{t-d}$ be the action-loss vector from d rounds earlier. Its usefulness depends on how far the losses have moved, which we measure by $E_2 = \sum_t \|c_t - m_t\|_\infty^2$ (Figure 1). Our lower bound asks how much regret this mismatch forces even when m_t is supplied for free.

In short, when the action-to-state map is known the number of actions stops setting the price (Theorem 1), but no algorithm escapes the cost of waiting, even one handed the stale losses outright (Theorem 2). An unknown map can be estimated, with a weaker guarantee (Appendix D), and adapting to overlap and drift at once remains open (Section 6).

2 Related work

We connect two largely separate lines of work. *Delayed bandits.* Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map; we do the reverse. Zimmert and Seldin (2020) give the optimal $\widetilde{O}(\sqrt{KT} + \sqrt{D \log K})$ for total delay D, and Esposito et al. (2023) allow an unknown, possibly adversarial action-to-state map; fixing that channel is what makes its overlap geometry measurable through v_t. *Stale predictions.* Bar-On and Mansour (2025) bound regret by the inaccuracy of evolving observations, whereas we ask what delay-dependent cost is unavoidable; Wang et al. (2026) bound gradient prediction error without delay, and Zhang et al. (2026) share the motivation in a stochastic Bayesian model, whereas ours is adversarial. *Mediator feedback* is the closest structural connection: $v_t - 1$ is the χ^2 mutual information of Eldowa et al. (2024). Our contribution is not that quantity but its role when the outcome at the observed state is delayed, and the separation of that benefit from an independent temporal obstruction (Appendix E). Feedback graphs and off-policy coverage (Gabbianelli et al., 2023) are related forms of information sharing; Appendix A compares them.

3 Problem formulation

There are T rounds, K actions, a finite state space $\mathcal{S}$, a known action-to-state matrix P, and a fixed known delay d; all logarithms are natural and $\widetilde{O}$ hides logarithmic factors. Every round follows the chain $A_t \xrightarrow{P} S_t \xrightarrow{\theta_t} X_t$: the learner draws A_t from a distribution x_t, observes $S_t \sim P(\cdot \mid A_t)$ at once, and receives the delayed outcome only at time $t + d$, encoded as a loss $X_t \in [0, 1]$ so that smaller is better, with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$, where $\mathcal{F}_{t-1}$ is everything seen before round t; outcomes are conditionally independent across rounds given the states. The conditional mean depends on A_t only through S_t. This *state-sufficiency* assumption licenses cross-action pooling: an outcome observed at state S_t may update every action that could have produced that state. If an action affects the outcome directly, in a way not represented in S_t, the argument no longer applies. Here x_t depends only on $\mathcal{F}_{t-1}$; X_r may first affect the action distribution at round $r+d+1$, equivalently the round-t decision uses outcomes only from rounds $r \le t-d-1$.

The matrix $P \in [0,1]^{K\times|\mathcal{S}|}$ is row-stochastic and fixed, and the *state-loss means* $\theta_t \in [0,1]^{|\mathcal{S}|}$ are *oblivious*: chosen in advance, not reacting to the learner. Action losses are $c_t = P\theta_t$ and performance is the static *pseudo-regret*, the gap to the best single action in hindsight, $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$. **Oracle for the lower bound.** For Theorem 2 only, the learner also receives the *stale action-loss vector* $m_t \triangleq c_{t-d}$ before choosing A_t, with $m_t = c_1$ for $t \le d$. This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading m_t is by $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \le T$, which stays small when errors are small even if frequent. The geometry of P thus governs sharing across actions and the evolution of θ_t governs staleness across time, the two axes of Section 4.

4 What the theory guarantees

Two results follow: the number of actions stops setting the price, for the algorithm one would actually run, and no sharing removes the cost of waiting. A rotating variant and one for merged states are stated and proved in Appendix C.

When an outcome arrives, do not update only the action played: update every action by how likely it was to produce the observed state. Here P is known and m_t is not given. For each state $s \in \mathcal{S}$ let $q_t(s) = \sum_{a=1}^K x_t(a)P(s \mid a)$, and give each action $a \in \{1,\dots,K\}$ the estimate $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$. It is correct on average, and how noisy it is depends on how much the actions overlap, through the *effective dimension* $v_t = \sum_s q_t(s)^{-1}\sum_a x_t(a)P(s \mid a)^2 \in [1, |\mathcal{S}|]$. STATE-EXP3 runs m copies of EXP3 (Auer et al., 2002) in rotation at learning rate η, copy $i \in \{0,\dots,m-1\}$ taking every mth round, and adds $\hat{c}_t$ to a copy's running total once the outcome is usable. A round costs $O(K)$ (Appendix B, with pseudocode and the χ^2 reading of v_t).

Theorem 1. *Let P be known and (θ_t) oblivious, and pick any $V^- \ge \sum_t (v_t - 1)$ before the run. Then* STATE-EXP3 *with $m = 1$ and any $\eta \le 1/(e(d+1))$ satisfies $\mathbb{E}[R_T] \le \sqrt{2(3.3V^- + 2dT)\log K} + e(d+1)\log K + d$.*

So K is replaced by V^-, which counts how differently the actions behave rather than how many there are, and the delay enters additively at $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$. It covers the algorithm every experiment below runs, at rates those runs already satisfy. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V^- = (\bar{v} - 1)T$ is admissible and depends on P alone. Rotating $d+1$ copies gives $\widetilde{O}(\sqrt{(d+1)V\log K})$ and merging similar states lowers V again at an explicit bias (Appendix C); none of the three adapts to E_2.

Where this stops. How much does waiting cost even when the stale losses are free? Theorem 2 shows a cost from time alone, not a joint overlap-drift bound: the construction gives each of the J drifting directions a private state, removing useful pooling and forcing $J \le |\mathcal{S}| - 1$ (Appendix F).

Theorem 2. *Fix T, a delay $1 \le d \le T/2$, a drift budget $\mathcal{E} \le T/4$, and $1 \le J \le \min\{K-1, |\mathcal{S}|-1\}$. Every algorithm, even one handed the exact losses from d rounds ago, meets some instance fixed in advance with $E_2 \le \mathcal{E}$ on which $\mathbb{E}[R_T] = \Omega(\sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$. Without that help, and if $K \le T$, the best regret any algorithm can guarantee is $\Omega(\sqrt{KT} + \sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$, up to constants.*

The first term is the usual cost of exploring. The second is the cost of delay: it grows with the wait, how far the losses move during it, and the number J of directions the best fixed action can use. They come from different hard instances, so adding them overstates the truth by at most a factor of two.

5 Experiments

The claim is about instances, so it must be checked where the number of actions and the number of distinct behaviours come apart. With $K = 200$ generated items and six levels of engagement the estimated dimension is 1.97, and the state-pooled variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at $d = 10, 50, 200$. That setting is hand-structured, not fitted to data, so it shows the mechanism, not a deployed system. An online plug-in $\hat{P}_t$ stays within 0.5 per cent of known P (study 2).

The advantage widens with the menu: at $|\mathcal{S}| = 4$ the single copy beats action-level weighting on 73, 99 and 100 of 100 seeds at $K = 8, 40, 80$, the paired gain rising from 19.3 ± 3.4 to 169.3 ± 9.9 (study 1).

It also beats the minimax-optimal delayed-bandit method of Zimmert and Seldin (2020), tuned over a grid under the reported tuning protocol.

d	action-level	tuned (Zimmert and Seldin, 2020)	STATE-EXP3	cut	cut
10	4577	2959	949	79.3%	67.9%
50	4752	3557	1728	63.6%	51.4%
200	5307	4735	3245	38.9%	31.5%

Funnel family, $K = 200$, $|\mathcal{S}| = 6$, $T = 20000$, 8 paired seeds; cuts are against action-level weighting and against the tuned baseline (study 16).

These runs use the single-copy algorithm, the one Theorem 1 covers; Appendix H also reports the rotating variant. A best-state rule leads seven of nine settings, but where its assumption fails its regret grows linearly and its spread across seeds is four times ours.

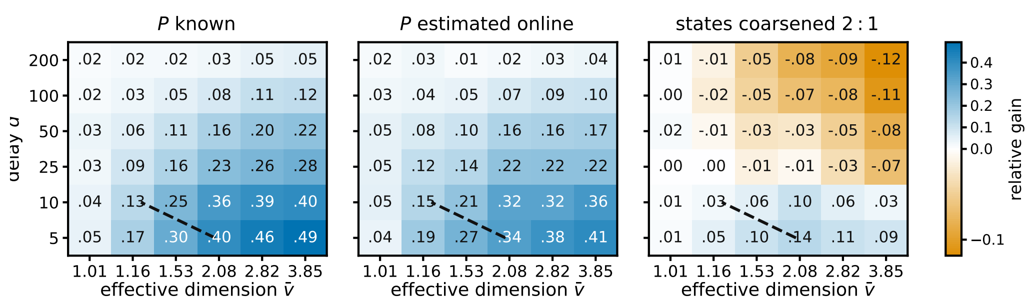

Figure 2: **Where pooling pays.** Relative gain $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$ of pooling through the state, single copy, $K = 40$, $|\mathcal{S}| = 8$, $T = 5000$, 12 paired seeds; blue is a gain. $\widehat{\bar{v}}$ estimates $\sup_x v(x)$ for the raw map by Monte Carlo, which for the coarsened panel is not the dimension of the coarsened map that Theorem 6 charges. The dashed line compares two upper bounds for the rotating variant, so it is a reference and not a prediction for the variant plotted.

Figure 2 sweeps row overlap against delay and marks where pooling pays, in the three settings the studies treat separately: P known, P estimated, and states coarsened.

6 Discussion

Pool when many actions lead to similar states, coarsen when grouped states agree, and distrust old information when what the states predict moves fast. Because P is known, the first is checkable before anything is deployed: compute $\bar{v}$ from the channel, and if it sits far below K, the state channel is worth using.

Theorem 2 says what no algorithm can do about the second: waiting leaves the learner uncertain about what the states now predict, and that uncertainty is irreducible, surviving even when the exact losses from d rounds ago are handed over free. Three limits bound the rest. The evidence is generated, not measured; $\bar{v}$ is estimated by sampling, so the guarantee invoked is that at $V = |\mathcal{S}|T$; and the two bounds lie on different axes, so adapting to both at once remains unproved.

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A Extended related work

Guinet et al. (2022) use the name *effective dimension* for an unrelated quantity, the number of directions a censored-feedback problem can resolve; it is not v_t and the observation model differs. The minimax characterisation of delayed adversarial bandits traces to Cesa-Bianchi et al. (2019), with Joulani et al. (2013) giving the additive form of the delay penalty in the stochastic case and Flaspohler et al. (2021) studying optimism under delay directly. Feedback graphs (Alon et al., 2015; Esposito et al., 2022) and off-policy coverage (Gabbianelli et al., 2023) share information across actions through a different channel: there the learner sees other actions' losses, whereas here it sees a state whose loss it must still estimate. Delay has also been studied across cooperating agents (Hsieh et al., 2022), and intermediate observations in a stochastic factored model (Mussi et al., 2024). On predictors rather than delay, Wei et al. (2020) bound regret in terms of a loss predictor's accuracy and Zhang (2026) give second-order path-length guarantees; both measure a prediction error as E_2 does, without the delay that makes it stale.

B The state-pooled estimator

The algorithm. Inputs are the action-to-state matrix P, the delay d, the number of copies m, and the rate $\eta > 0$. The analyzed setting is $m = d + 1$ and the practical variant is $m = 1$. Round r's outcome becomes usable at round $r + d + 1$, which for $m = d + 1$ is the next round belonging to the copy that played round r.

1. Set $L_i \leftarrow (0, \ldots, 0) \in \mathbb{R}^K$ for $i = 0, \ldots, m - 1$, and let $\mathcal{P}$ be an empty table of rounds awaiting their outcome.
2. For $t = 1, \ldots, T$:
   (a) *Deliver.* If $r \triangleq t - d - 1 \geq 1$, read $(i_r, q_r(S_r), S_r, X_r)$ from $\mathcal{P}$, remove it, and update only that copy, $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r \mid a) X_r / q_r(S_r)$ for every a. This is the pooled step, and it touches every action rather than the one played.
   (b) *Select the copy.* $i \leftarrow t \bmod m$.
   (c) *Play.* $x_t(a) \propto \exp(-\eta L_i(a))$, draw $A_t \sim x_t$, and observe S_t at once.
   (d) *Record.* Compute the scalar $q_t(S_t) = \sum_a x_t(a) P(S_t \mid a)$ and store $(i, q_t(S_t), S_t, \cdot)$ in $\mathcal{P}$, to be completed when X_t arrives at time $t + d$.

Steps (a), (c) and (d) are each $O(K)$ and there is no optimization subproblem, so a round costs $O(K)$ once P is stored. The state is held in the m vectors L_i, which is mK numbers, the matrix P, which is $K|\mathcal{S}|$ numbers, and four scalars for each of the at most d rounds in flight. Only the scalar $q_r(S_r)$ is needed from round r, not the vector x_r.

A worked example. Take three actions and two states, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \\ 0 & 1 \end{pmatrix}, \qquad x_t = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),$$

so $q_t(s_1) = \frac{1}{2}(0.8) + \frac{1}{4}(0.4) + \frac{1}{4}(0) = 0.5$ and $q_t(s_2) = 0.5$. The third action cannot produce s_1 at all. Suppose the learner draws a_1, observes s_1, and receives $X_t = 1$ at time $t + d$. Then $\hat{c}_t(a) = P(s_1 \mid a)/0.5$, so $\hat{c}_t = (1.6,\, 0.8,\, 0)$ against the action-level $\tilde{c}_t = (2, 0, 0)$. One observation updates all three actions: a_2 is charged although it was never played, because it had a 0.4 chance of producing the state seen, and a_3 is charged nothing, because it could not have produced that state. Nothing is biased by this. With $\theta_t = (1, 0)$ the true losses are $c_t = (0.8, 0.4, 0)$, and averaging $\hat{c}_t$ over $S_t \sim q_t$ returns exactly that. What is bought is the second moment: $v_t = 1.44$ here, against $K = 3$ for the action-level estimate, and with $\theta_t = (1, 1)$ both are attained exactly.

Remark 3 (The χ^2 form). Under x_t the pair (A_t, S_t) has law $x_t(a)P(s \mid a)$ with marginals x_t and q_t, so

$$\chi^2\big(P_{A_t,S_t} \,\big\|\, P_{A_t}\otimes P_{S_t}\big) = \sum_{a,s} \frac{\big(x_t(a)P(s \mid a) - x_t(a)q_t(s)\big)^2}{x_t(a)q_t(s)}$$
$$= \sum_{a,s} \frac{x_t(a)P(s \mid a)^2}{q_t(s)} - 2 + 1 \;=\; v_t - 1,$$

the cross term summing to $\sum_{a,s} x_t(a)P(s \mid a) = 1$ and the last to $\sum_{a,s} x_t(a)q_t(s) = 1$. So $v_t = 1$ exactly when A_t and S_t are independent, which pins the rows of P only on the support of x_t and therefore means all rows agree whenever x_t has full support, as exponential weights does. If P is deterministic and every state is reached by some action in the support of x_t, then v_t equals the number of states that support reaches. In particular, under full-support play and an onto deterministic P, $v_t = |\mathcal{S}|$. This is an identity, not a bound; it is used for reading v_t, and once in the proof of Theorem 1.

Proposition 4. *Fix t, let x_t be the play distribution, $q_t(s) = \sum_a x_t(a)P(s \mid a)$ the induced state distribution, and $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$, which is well defined because only states with $q_t(s) > 0$ occur. For statements involving $1/x_t(a)$, restrict to actions with $x_t(a) > 0$;* State-EXP3 *itself has full support on every action. Then $\hat{c}_t(a) \geq 0$, $\hat{c}_t(a) \leq 1/x_t(a)$, $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$, and*

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\leq\; v_t := \sum_{s:q_t(s)>0} \frac{\sum_a x_t(a)P(s \mid a)^2}{q_t(s)}, \qquad 1 \leq v_t \leq |\mathcal{S}|,$$

states of zero probability being omitted throughout, which is the convention $0/0 = 0$. The action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ has the same first three properties and weighted second moment at most K.

Proof. Nonnegativity is immediate from $X_t \geq 0$. Given $\mathcal{F}_{t-1}$ the state S_t has distribution q_t, and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$ by the model, since that conditional mean does not depend on the action, so $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s \mid a)\theta_t(s)/q_t(s) = c_t(a)$. For the range, $q_t(s) \geq x_t(a)P(s \mid a)$ for every a, so $P(s \mid a)/q_t(s) \leq 1/x_t(a)$ and $X_t \leq 1$. For the second moment, $X_t^2 \leq 1$ gives $\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq \sum_s P(s \mid a)^2/q_t(s)$, and exchanging the sums gives v_t. Each summand of v_t is at most 1, since $P(s \mid a) \leq 1$ makes $\sum_a x_t(a)P(s \mid a)^2 \leq q_t(s)$, and at least $q_t(s)$, since Jensen applied to $a \mapsto P(s \mid a)$ under x_t gives $\sum_a x_t(a)P(s \mid a)^2 \geq q_t(s)^2$. Summing the two bounds gives $1 \leq v_t \leq |\mathcal{S}|$. The action-level claims are the same computation with the state replaced by the played action, whose weighted second moment is $\sum_a x_t(a)/x_t(a) = K$. ☐

The two ends are attained. If every action induces the same state distribution then $v_t = \sum_s q_t(s) = 1$, and if P is deterministic, each action reaching a single state, then every $P(s \mid a)$ is 0 or 1 and each summand is exactly 1, so v_t counts the states reached and equals $|\mathcal{S}|$ when every state has an action reaching it. Disjoint row supports give $v_t = K$ rather than $|\mathcal{S}|$, since the summands over one action's private states add to $\sum_s P(s \mid a) = 1$, so a sensor with many private states per action resolves few of them. So v_t reads the overlap between the rows of P rather than their length, and $\bar{v} = \sup_x v(x)$, the supremum over play distributions, depends on P alone. An outcome observed at a state updates the estimate for every action that can reach it, which is why the construction of Section 4 gives each drifting coordinate a private state, forcing $v_t = |\mathcal{S}|$ there. The immediately observed pairs (A_t, S_t) identify P separately, though Theorem 5 assumes it known.

C Proofs: pooling and grouping

Two results are stated here rather than in the body, since Theorem 1 covers the algorithm actually run. The first is the interleaved variant.

Theorem 5. *Let P be known and (θ_t) oblivious. Pick before the run any fixed number V with $\sum_t v_t \leq V$ always. Then* State-EXP3 *with $m = d + 1$ and the matching learning rate satisfies $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V \log K}$. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V = \bar{v}T \leq |\mathcal{S}|T$ is one such choice, and it depends on P alone.*

The second merges states that are not worth telling apart. Group them with a map g from $\mathcal{S}$ onto a smaller set $\mathcal{Z}$ and run the same algorithm on $g(S_t)$, where $P^g(z \mid a) = \sum_{s:g(s)=z} P(s \mid a)$. Grouping never raises the dimension, but the estimate now tracks a group average, so the widest spread of losses inside a group, $\delta_t(g) = \max_z \max_{s,s':g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$, is the error introduced.

Theorem 6. *Under the hypotheses of Theorem 5, running on g with $\sum_t v_t(g) \leq V(g)$ gives $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g)\log K} + 2\sum_t \delta_t(g)$.*

Merging need not preserve state-sufficiency, so the grouped estimate is right not for c_t but for a stand-in loss differing from it by at most $\delta_t(g)$ on every action. Keeping states apart preserves differences that matter but learns more slowly; merging shares more but pays $2\sum_t \delta_t(g)$. The best grouping is the true one: with 16 states from four hidden groups, sizes $1, 2, 4, 8, 16$ give regret 1590, 831, 754, 920, 1080, lowest at four, which no algorithm was told. The map g is fixed in advance; learning it from data remains open.

Proof idea. The pooled estimator is unbiased for every action, while Proposition 4 bounds its play-weighted second moment by v_t rather than K. With $m = d + 1$ interleaved copies, the outcome generated on a round belonging to one copy is available before that copy acts again, so each copy is an ordinary exponential-weights process with immediate feedback. Summing the $d + 1$ potential inequalities gives $(d + 1)\log K/\eta$, and the pooled second moments give $\frac{\eta}{2}\sum_t v_t$.

Proof of Theorem 5. Fix a copy i and let T_i be the number of rounds it owns, so $\sum_i T_i = T$. Round t's outcome is usable from round $t + d + 1$, which with $m = d + 1$ is exactly $t + m$, the copy's next round, so within a copy the feedback is undelayed and L_i carries exactly the estimates of that copy's earlier rounds. Exponential weights on nonnegative losses give, for any action $a^\star$ and any $\eta > 0$,

$$\sum_{t\in i}\big(\langle x_t, \hat{c}_t\rangle - \hat{c}_t(a^\star)\big) \leq \frac{\log K}{\eta} + \frac{\eta}{2}\sum_{t\in i}\sum_a x_t(a)\hat{c}_t(a)^2,$$

by the usual potential telescoping with $e^{-z} \leq 1 - z + z^2/2$, which needs $z \geq 0$ and therefore $\hat{c}_t \geq 0$, supplied by Proposition 4. Each x_t is $\mathcal{F}_{t-1}$-measurable, because L_i then holds only rounds $r \leq t - m = t - d - 1$, so taking expectations turns the left side into $\sum_{t\in i}(\langle x_t, c_t\rangle - c_t(a^\star))$ by unbiasedness and the quadratic term into at most $\mathbb{E}[\sum_{t\in i} v_t]$, again by Proposition 4. Summing over the $m = d+1$ copies and using $\sum_i \min_a \sum_{t\in i} c_t(a) \leq \min_a \sum_t c_t(a)$, which is what lets the copies be compared against one common action, gives $\mathbb{E}[R_T] \leq (d+1)\log K/\eta + \frac{\eta}{2}\mathbb{E}[\sum_t v_t]$. If $\sum_t v_t \leq V$ surely then $\eta = \sqrt{2(d+1)\log K/V}$ balances the two terms and gives $\sqrt{2(d+1)V\log K}$, and $V = \bar{v}T$ is admissible by the definition of $\bar{v}$. $\square$

The delay enters as a factor because the copies do not share their estimates, each paying $\log K/\eta$ over $T/(d + 1)$ rounds. Sharing them is exactly the $m = 1$ algorithm of the open problem of Section 6.

Lemma 7 (Coarsening)**.** *For every play distribution and every g, $v_t(g) \leq v_t$.*

Proof. It suffices to merge two states, the general case following by induction on the merges, and a merged group of zero probability contributes nothing on either side by the convention above. Write P_1, P_2 for the two columns, $A_j = \sum_a x_t(a)P_j(a)^2$, $B_j = \sum_a x_t(a)P_j(a)$ and $\lambda = B_1/(B_1 + B_2)$. By Cauchy–Schwarz, $(P_1+P_2)^2 \leq \lambda^{-1}P_1^2+(1-\lambda)^{-1}P_2^2$ pointwise in a, so $\sum_a x_t(a)(P_1+P_2)^2 \leq (B_1 + B_2)(A_1/B_1 + A_2/B_2)$, and dividing by the merged denominator $B_1 + B_2$ shows the merged summand is at most $A_1/B_1 + A_2/B_2$, the two it replaces. $\square$

Proof of Theorem 6. Running the algorithm on g is running it on the sensor $(\mathcal{Z}, P^g)$. The surrogate losses below depend on x_t and so are predictable rather than oblivious, but the proof of Theorem 5 uses obliviousness nowhere, only that each c_t^g is $\mathcal{F}_{t-1}$-measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses $c_t^g(a) = \sum_z P^g(z \mid a)\bar{\theta}_t(z)$, where $\bar{\theta}_t(z) = \sum_{s:g(s)=z} q_t(s)\theta_t(s)/q_t^g(z)$ is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 4 with S_t replaced by $g(S_t)$. Fix a and z. Both $\{q_t(s)/q_t^g(z)\}$ and $\{P(s \mid a)/P^g(z \mid a)\}$ are probability vectors on $\{s : g(s) = z\}$, so their θ_t-averages lie in a common interval of length $\delta_t(g)$ and differ by at most $\delta_t(g)$. Weighting by $P^g(z \mid a)$ and summing over z gives $|c_t^g(a) - c_t(a)| \le \delta_t(g)$ for every a, hence $\langle x_t, c_t\rangle - c_t(a^\star) \le \langle x_t, c_t^g\rangle - c_t^g(a^\star) + 2\delta_t(g)$. Summing over t adds $2\sum_t \delta_t(g)$, and Lemma 7 makes $V(g)$ no larger than a bound valid for the raw states. $\square$

D An unknown action-to-state map

The plug-in estimator. The pairs (A_t, S_t) arrive without delay, so P is estimable from the learner's own play while the outcomes are still in flight. Let $N_a(t)$ count the rounds before t that played a and $N_{a,s}(t)$ those that also produced s, and set $\hat{P}_t(s \mid a) = (N_{a,s}(t) + \alpha)/(N_a(t) + \alpha|\mathcal{S}|)$ for a smoothing constant $\alpha \in (0, 1]$ fixed independently of T, which is row-stochastic with every entry positive, so $\hat{q}_t = x_t^\top \hat{P}_t$ is positive too, and both are $\mathcal{F}_{t-1}$-measurable. Smoothing moves an entry from the empirical row by at most $\alpha(|\mathcal{S}| - 1)/(N_a(t) + \alpha|\mathcal{S}|)$, attained when every observation of a sits on one state, and moves the row in ℓ_1 by at most twice that. Both are of lower order than the sampling term at fixed α and must be carried otherwise, and Appendix H shows the choice matters in practice. The plug-in estimate is $\hat{c}_t(a) = \hat{P}_t(S_t \mid a)X_t/\hat{q}_t(S_t)$. Write $\varepsilon_t = \max_a \|\hat{P}_t(\cdot \mid a) - P(\cdot \mid a)\|_1$ for its accuracy and $\rho_t = \max_{s: q_t(s)>0} |\hat{q}_t(s) - q_t(s)|/q_t(s)$ for the relative error it induces on the state distribution, states of zero probability under the play being omitted since the estimator never charges them.

Proposition 8. *On $\{\rho_t \le \frac{1}{2}\}$ the plug-in estimate satisfies $\hat{c}_t \ge 0$ and $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \le 2\hat{v}_t \le 2|\mathcal{S}|$, and its conditional mean $\bar{c}_t$ satisfies $\sum_a x_t(a)(\bar{c}_t(a) - c_t(a)) = 0$ exactly, for every $\hat{P}_t$ and with no hypothesis at all, and $|\bar{c}_t(a) - c_t(a)| \le 2\varepsilon_t(a) + 2\rho_t$ for every a, where $\varepsilon_t(a) = \|\hat{P}_t(\cdot \mid a) - P(\cdot \mid a)\|_1$.*

Proof. Write $\Delta_t = \hat{P}_t - P$ and $\delta_t = \hat{q}_t - q_t = \sum_a x_t(a)\Delta_t(\cdot \mid a)$. Substituting these into $\bar{c}_t(a) = \sum_s q_t(s)\hat{P}_t(s \mid a)\theta_t(s)/\hat{q}_t(s)$ and cancelling gives $\bar{c}_t(a) - c_t(a) = \sum_s \theta_t(s)[q_t(s)\Delta_t(s \mid a) - P(s \mid a)\delta_t(s)]/\hat{q}_t(s)$, so with $\theta_t \in [0,1]^{|\mathcal{S}|}$,

$$|\bar{c}_t(a) - c_t(a)| \;\le\; \sum_s \frac{q_t(s)|\Delta_t(s \mid a)| + P(s \mid a)|\delta_t(s)|}{\hat{q}_t(s)}.$$

The play-weighted identity needs no hypothesis. By definition $\sum_a x_t(a)\hat{P}_t(s \mid a) = \hat{q}_t(s)$, so $\sum_a x_t(a)\hat{c}_t(a) = X_t$ whatever $\hat{P}_t$ is, and taking conditional expectations gives $\sum_a x_t(a)\bar{c}_t(a) = \sum_s q_t(s)\theta_t(s) = \langle x_t, c_t\rangle$. On $\{\rho_t \le \frac{1}{2}\}$ we have $q_t \le 2\hat{q}_t$, and for the per-action bias the first sum of the display is at most $2\|\Delta_t(\cdot \mid a)\|_1 = 2\varepsilon_t(a)$ and the second at most $2\sum_s P(s \mid a)|\delta_t(s)|/q_t(s) \le 2\rho_t$, because $P(\cdot \mid a)$ is a probability vector. The play-weighted identity does *not* survive reweighting: at $x_t = (\frac{1}{2}, \frac{1}{2})$, $P = I$, $\hat{P} = \left(\begin{smallmatrix} .9 & .1 \\ .2 & .8 \end{smallmatrix}\right)$ and $\theta_t = (0, 1)$ the bias is $(\frac{1}{9}, -\frac{1}{9})$ with zero x_t-average, while weights $(1, 0)$ leave $\frac{1}{18}$. For the second moment, $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \le \sum_s \frac{q_t(s)}{\hat{q}_t(s)} \cdot \frac{\sum_a x_t(a)\hat{P}_t(s|a)^2}{\hat{q}_t(s)} \le 2\hat{v}_t$, and $\hat{v}_t \le |\mathcal{S}|$ by Proposition 4 applied to $\hat{P}_t$. $\square$

The warm-start guarantee. WARM-START STATE-EXP3 plays uniformly for n_0 rounds, forms $\hat{P} = \hat{P}_{n_0+1}$ from those rounds alone, discards the cumulative estimates together with every outcome of a warm-start round that has not yet arrived, and then runs STATE-EXP3 with $m = d + 1$, that frozen $\hat{P}$, and each play mixed with the uniform action at rate γ. Freezing makes $\hat{P}$ measurable with respect to the first phase, so ε and ρ are fixed before the second phase begins, and discarding is what keeps the first phase's inaccurate updates out of every later distribution.

Theorem 9. *Suppose $\bar{p}(s) \ge 1/(\kappa|\mathcal{S}|)$ for every s, fix $0 < \gamma \le \frac{1}{2}$ and $1 \le d \le T/2$, and take $n_0 \ge c\,\gamma^{-2}K\kappa^2|\mathcal{S}|^2\log(K|\mathcal{S}|T)$ for a universal c. Then* WARM-START STATE-EXP3 *satisfies, for every $\eta > 0$,*

$$\mathbb{E}[R_T] \;\le\; n_0 + 1 + \gamma T + \frac{(d+1)\log K}{\eta} + 2\eta|\mathcal{S}|T + T\big(4\bar{\varepsilon} + 4\bar{\rho}\big),$$

where $\bar\varepsilon = \widetilde{O}(\sqrt{K|\mathcal{S}|/n_0})$ and $\bar\rho = \widetilde{O}(\kappa|\mathcal{S}|\sqrt{K/n_0}/\gamma) \le \frac{1}{2}$ bound ε and ρ on an event of probability at least $1 - 1/T$. Tuning η and then $\gamma \asymp (\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4}$ and $n_0 \asymp T^{4/5}(\kappa|\mathcal{S}|)^{2/5}K^{1/5}$ gives $\mathbb{E}[R_T] = \widetilde{O}\big(\sqrt{d\,|\mathcal{S}|T\log K} + (\kappa|\mathcal{S}|)^{2/5}K^{1/5}T^{4/5}\big)$, provided $T = \widetilde{\Omega}(K(\kappa|\mathcal{S}|)^2)$. That condition constrains T against K, $|\mathcal{S}|$ and κ only, so the hypothesis $d \le T/2$ is what carries the delay. Together they give $n_0 \le T/2 \le T - d$, and the same condition supplies the lower bound above with some $\gamma \le \frac{1}{2}$. Outside that regime the tuning is vacuous and $R_T \le T$ is the statement.

Proof. The first phase has losses in $[0,1]$, so it contributes at most n_0. Let G be the event that $N_a(n_0+1) \ge n_0/(2K)$ for every a, that $\|\hat{P}(\cdot \mid a) - P(\cdot \mid a)\|_1 \le \bar\varepsilon$ for every a, and that $\max_{s,a}|\hat{P}(s \mid a) - P(s \mid a)| \le \bar\rho\gamma/(\kappa|\mathcal{S}|)$. Uniform play gives $\mathbb{E}[N_a] = n_0/K$, so a multiplicative Chernoff bound, the ℓ_1 deviation bound for a multinomial, Hoeffding, and a union bound over the $K|\mathcal{S}|$ entries put $\Pr(G^c) \le 1/T$ at the stated n_0, the pseudo-counts adding $2\alpha(|\mathcal{S}|-1)/(N_a+\alpha|\mathcal{S}|)$ to each ℓ_1 deviation and half that to each entry, of lower order at fixed α. Off G bound the second phase by T, contributing at most 1.

On G the quantity $\hat{P}$ is measurable with respect to the first phase, and mixing gives $q_t(s) \ge \gamma\bar{p}(s) \ge \gamma/(\kappa|\mathcal{S}|)$ at every later round, so $\rho_t \le \kappa|\mathcal{S}|\max_{s,a}|\hat{P}(s \mid a) - P(s \mid a)|/\gamma \le \bar\rho \le \frac{1}{2}$ for every $t > n_0$ and every realization of the second phase. Proposition 8 therefore applies at every such round, and conditionally on the first phase the second is exactly the algorithm of Theorem 5 driven by a nonnegative estimate with weighted second moment at most $2\hat{v}_t \le 2|\mathcal{S}|$. Its proof gives $\mathbb{E}[\sum_{t>n_0}(\langle p_t, \bar{c}_t\rangle - \bar{c}_t(a^\star))] \le (d+1)\log K/\eta + 2\eta|\mathcal{S}|T$, the extra factor of two coming from $p_t \le x_t/(1-\gamma) \le 2x_t$. Playing x_t rather than p_t costs at most γ per round since $c_t \in [0,1]^K$, so it remains to pass from $\bar{c}_t$ to c_t under p_t. Proposition 8 gives $\langle x_t, b_t\rangle = 0$ for the per-action bias b_t, and $x_t = (1-\gamma)p_t + \gamma u$ for the uniform u, so $\langle p_t, b_t\rangle = -\frac{\gamma}{1-\gamma}\langle u, b_t\rangle$, which for $\gamma \le \frac{1}{2}$ is at most $\max_a|b_t(a)|$ in absolute value. Each round therefore adds $|\langle p_t, b_t\rangle| + |b_t(a^\star)| \le 2\max_a|b_t(a)| \le 4\bar\varepsilon + 4\bar\rho$. Collecting the four contributions gives the display. For the rate, balancing γT against $2T\bar\rho$ gives the stated γ and a combined $\widetilde{O}(T(\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4})$, and balancing that against n_0 gives the stated n_0, under which $T\bar\varepsilon$ and the constraint on n_0 are of lower order. □

Safe blending. Blending the plug-in with the action-level estimate makes the guarantee safe. Let $\lambda_t \in [0,1]^K$ be $\mathcal{F}_{t-1}$-measurable and put $\breve{g}_t(a) = (1-\lambda_t(a))\tilde{c}_t(a) + \lambda_t(a)\hat{c}_t(a)$.

Proposition 10. *$\breve{g}_t \ge 0$, its conditional mean satisfies $\mathbb{E}[\breve{g}_t(a) \mid \mathcal{F}_{t-1}] - c_t(a) = \lambda_t(a)\,(\bar{c}_t(a) - c_t(a))$ for the plug-in's conditional mean $\bar{c}_t$ of Proposition 8, and*

$$\sum_a x_t(a)\mathbb{E}[\breve{g}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\le\; (K - \Pi_t) + \hat{V}_t,$$

for $\Pi_t = \sum_a \lambda_t(a)$ and $\hat{V}_t = \sum_a \lambda_t(a)x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}]$, which satisfies $\hat{V}_t \le 2\hat{v}_t$ on $\{\rho_t \le \frac{1}{2}\}$ and not in general, since the step $q_t/\hat{q}_t \le 2$ behind it is exactly that event. We write $V_t^{\mathrm{ub}}(\tau)$ below for a computable upper bound on $\hat{V}_t$, reserving $\hat{V}_t$ for the conditional expectation itself.

Proof. Both parts are nonnegative and $\lambda_t(a) \in [0,1]$. The mean identity is linearity together with $\mathbb{E}[\tilde{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$. For the second moment, $u \mapsto u^2$ is convex, so $\breve{g}_t(a)^2 \le (1-\lambda_t(a))\tilde{c}_t(a)^2 + \lambda_t(a)\hat{c}_t(a)^2$, and $x_t(a)\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \le 1$ by Proposition 4. □

So a weight of $\lambda_t(a)$ on the plug-in buys a variance saving of $\lambda_t(a)$ and pays a bias of $\lambda_t(a)$ times the plug-in's, and both endpoints are exact. Let $r_t(a)$ and $\hat\rho_t$ be any sequences computable from the immediate pairs that are *simultaneously valid*, meaning $\varepsilon_t(a) \le r_t(a)$ for every a and $\rho_t \le \hat\rho_t$ at every $t \le T$, on an event of probability at least $1 - 1/T$. Write $V_t^{\mathrm{ub}}(\tau) = 2\sum_{a:r_t(a)\le\tau} x_t(a)\sum_s \hat{P}_t(s \mid a)^2/\hat{q}_t(s)$, which the learner can evaluate and which dominates $\hat{V}_t$ on $\{\rho_t \le \frac{1}{2}\}$. SAFE-POOL STATE-EXP3 sets $\lambda_t \equiv 0$ when $\hat\rho_t > \frac{1}{2}$, and otherwise picks the best member of the candidate set $\{\perp\} \cup \{r_t(a) : a \le K\}$, where $\perp$ stands for $\lambda_t \equiv 0$ and a numerical candidate τ stands for $\lambda_t(a) = \mathbf{1}\{r_t(a) \le \tau\}$, best meaning smallest

$$\Psi_t(\tau) = \tfrac{\eta}{2}\big[(K - \Pi_t(\tau)) + V_t^{\mathrm{ub}}(\tau)\big] + \big(4\tau + 4\hat\rho_t\big)\mathbf{1}\{\Pi_t(\tau) > 0\}, \qquad \Psi_t(\perp) = \tfrac{\eta}{2}K.$$

Write $\Psi_t^\star$ for that smallest value. The candidate $\perp$ is listed apart from the numerical thresholds because a valid radius may equal zero, in which case $\tau = 0$ pools the actions attaining it rather than abstaining, so $\perp$ rather than $\tau = 0$ is what keeps $\frac{\eta}{2}K$ available on every round.

Theorem 11. *For every $\eta > 0$,* Safe-Pool State-EXP3 *with $m = d+1$ satisfies*

$$\mathbb{E}[R_T] \;\le\; 1 + \frac{(d+1)\log K}{\eta} + \mathbb{E}\Big[\sum_{t=1}^{T} \Psi_t^\star\Big],$$

so $\eta = \sqrt{2(d+1)\log K/(KT)}$ gives $\mathbb{E}[R_T] \le 1 + \sqrt{2(d+1)KT\log K}$ on every instance, which is the guarantee of action-level exponential weights, with an improvement on each round whose $\Psi_t^\star$ is strictly below $\frac{\eta}{2}K$. The certificate is computed from the learner's own observations and so is random, which is why the display carries an expectation. The tuned bound does not, because $\Psi_t^\star \le \frac{\eta}{2}K$ holds on every path.

Proof. On the validity event, of probability at least $1-1/T$ and contributing at most 1 off it, pooling happens only when $\hat\rho_t \le \frac{1}{2}$ and hence only when $\rho_t \le \frac{1}{2}$, which is what makes V_t^{ub} dominate the conditional second moment. Proposition 10 then supplies a nonnegative estimate whose weighted second moment is at most $(K - \Pi_t) + \hat V_t$, so the argument of Appendix C applies to the surrogate losses. Passing to c_t adds both the played bias $|\sum_a x_t(a)\lambda_t(a)b_t(a)|$ and the comparator bias $\lambda_t(a^\star)|b_t(a^\star)|$, for $b_t = \bar c_t - c_t$. The cancellation of Proposition 8 is lost under per-action weights, as the counterexample there shows, so each is bounded separately. On a pooled action $\varepsilon_t(a) \le r_t(a) \le \tau$ gives $|b_t(a)| \le 2\tau + 2\hat\rho_t$, and since $\sum_a x_t(a)\lambda_t(a) \le 1$ and $\lambda_t(a^\star) \le 1$ the two together are at most $4\tau + 4\hat\rho_t$, charged only when something is pooled. The second-moment bound and the two bias terms are all functions of the observations, so the per-round costs they assemble into are the random $\Psi_t^\star$ and the sum is taken in expectation. For the tuned bound, $\perp$ belongs to the candidate set and $\Psi_t(\perp) = \frac{\eta}{2}K$, so $\Psi_t^\star \le \frac{\eta}{2}K$ on every path whatever the radii are. ☐

E Relation to other inaccuracy measures

Write $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$ for the inaccuracy measure of Bar-On and Mansour (2025). Unlike E_2, which charges only the worst action in a round, Λ_2 grows with how many actions are predicted wrongly, and since $\|v\|_\infty \le \|v\|_2 \le \sqrt{J}\|v\|_\infty$ for a vector v with J nonzero coordinates,

$$E_2 \;\le\; \Lambda_2 \;\le\; J\,E_2,$$

with both ends attained: the left when a single action is mispredicted, the right when all J are mispredicted equally. So a bound in E_2 is never weaker and can be J times stronger, which is why Theorem 2 is stated in E_2. The two measures agree exactly when the prediction error is concentrated on one action.

F Proofs: the lower bound

The measure E_2 is controlled by the state-loss variation W, and no converse bound holds in general.

Lemma 12 (Delay window)**.** *$E_2 \le d^2 W$ for $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$. The ratio $E_2/(d^2W)$ tends to one when θ drifts monotonically in one coordinate at a constant rate and $T/d \to \infty$, the first d rounds contributing $\sum_{j<d} j^2$ rather than d^2 each under the convention $m_t = c_1$.*

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in W alone is available.

Proof of Lemma 12. Throughout write $\Delta_j = \theta_j - \theta_{j-1}$, set $\theta_0 = \theta_1$, so $\Delta_1 = 0$, and read c_r as c_1 for $r \le 0$, matching the convention $m_t = c_1$ for $t \le d$ of Section 3. Then $c_t - c_{t-d} = P\sum_{j=t-d+1}^{t}\Delta_j$. Because the rows of P are probability vectors, averaging cannot increase the supremum norm, so $\|c_t - c_{t-d}\|_\infty \le \sum_j \|\Delta_j\|_\infty$. By Cauchy–Schwarz applied to the d summands, $\|c_t - c_{t-d}\|_\infty^2 \le d\sum_{j=t-d+1}^{t}\|\Delta_j\|_\infty^2$. Summing over t and using that each increment belongs to at most d windows gives $E_2 \le d^2 W$. The ratio $E_2/(d^2W)$ approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and P passes the increment through undiminished. Without the second condition the drift can lie in the kernel of P and the ratio can be zero. $\square$

Lemma 13 (Information isolation)**.** *If $h \le d$ then A_t is independent of the signs governing its own block, and this remains true when the learner is provided with the exact stale action-loss vector m_t.*

Proof of Lemma 13. For t in block b, every realized outcome usable when choosing A_t comes from a round $r \le t-d-1$. Since $t \le bh$ and $h \le d$, we have $r \le bh-d-1 < (b-1)h$, so every usable realized outcome depends only on signs from blocks strictly before b, whose joint law depends only on $\sigma_2, \dots, \sigma_{b-1}$. Hence by conditional independence A_t is independent of block b's signs. For the oracle, take any $h \le d$. If $t > d$ then $t-d \le bh-d \le (b-1)h$ by the same computation, so $m_t = c_{t-d}$ is measurable with respect to the signs of blocks strictly before b. If $t \le d$ then $m_t = c_1 = \frac{1}{2}\mathbf{1}$, which is deterministic because block 1 is neutral. In both cases m_t is independent of block b's signs, so adjoining it to the learner's information leaves the conclusion intact. $\square$

The construction in full. Take states $s_0, \dots, s_q$, a stationary deterministic map sending a_j to s_j for $j \le J$ and every other action to s_0, and independent Rademacher signs $\sigma_b^{(j)}$ for $2 \le b \le B = \lfloor T/h \rfloor$. Set $\theta_t(s_0) = \frac{1}{2}$ throughout, $\theta_t \equiv \frac{1}{2}$ on block 1, and $\theta_t(s_j) = \frac{1}{2} - \varepsilon\sigma_b^{(j)}$ on each later block b. The neutral first block makes $c_1 = \frac{1}{2}\mathbf{1}$, so the convention $m_t = c_1$ for $t \le d$ of Section 3 supplies a constant over exactly the rounds that precede any usable feedback, which is what Lemma 13 needs at $b = 1$. Rounds after the last complete block carry no drift, costing a constant factor.

Lemma 14 (Rademacher lower tail)**.** *Let S_B be a sum of $B \ge 32$ independent Rademacher signs. For every $1 \le u \le \sqrt{B}/32$, $\Pr(S_B \ge u\sqrt{B}) \ge \frac{u}{4}e^{-64u^2}$.*

Proof. Write $S_B = 2X - B$ with $X \sim \mathrm{Bin}(B, \frac{1}{2})$, so that $\{S_B \ge u\sqrt{B}\} = \{X \ge B/2 + u\sqrt{B}/2\}$. Passing to X removes the lattice gap, since X takes every integer value in $[0, B]$ whereas S_B takes only those of the parity of B. Put $n = \lfloor B/2 \rfloor$ and $t = \lceil u\sqrt{B}/2 \rceil + 1$, so $X \ge n + t$ implies the event, and $t \le u\sqrt{B}$ because $u\sqrt{B} \ge \sqrt{32} > 4$.

For $j \ge 1$ the ratio of consecutive binomial probabilities is $\Pr(X = n+j)/\Pr(X = n+j-1) = (B-n-j+1)/(n+j)$. Using $(B-1)/2 \le n \le B/2$, its distance from one is at most $(2j-1)/(n+j) \le (4j-2)/(B-1) \le 8j/B$ for $B \ge 2$. Restricting to $j \le B/16$ keeps $8j/B \le \frac{1}{2}$, where $\log(1-x) \ge -2x$ holds, so summing over $i \le j$ gives $\Pr(X = n+j) \ge \Pr(X = n)e^{-16j(j+1)/(2B)} \ge \Pr(X = n)e^{-16j^2/B}$. The central term satisfies $\Pr(X = n) \ge 1/(2\sqrt{B})$ for every $B \ge 2$.

Sum over the t integers $j \in \{t, \dots, 2t-1\}$, all of which are at most $2t$ and hence at most $B/16$ because $u \le \sqrt{B}/32$ forces $2t \le 2u\sqrt{B} \le B/16$. Each term is at least $e^{-64t^2/B}/(2\sqrt{B})$, and $t \le u\sqrt{B}$ makes the exponent at most $64u^2$, so $\Pr(X \ge n+t) \ge t\,e^{-64u^2}/(2\sqrt{B}) \ge \frac{u}{4}e^{-64u^2}$ using $t \ge u\sqrt{B}/2$. $\square$

Three regimes give the maximal inequality the lower bound needs. Each $S_B^{(j)}$ is sub-Gaussian with variance proxy B, so $\mathbb{E}\max_j(S_B^{(j)})^+ \le \sqrt{2B\log J} + \sqrt{B}$ throughout. For the matching lower bound, first suppose $\log J \le B/8$ and $\log J \ge 128$. Taking $u = \sqrt{\log J/128}$, which then lies in $[1, \sqrt{B}/32]$, makes $e^{-64u^2} = J^{-1/2}$, so $\Pr(S_B^{(j)} \ge u\sqrt{B}) \ge \frac{u}{4}J^{-1/2}$ and independence across j gives $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \le e^{-u\sqrt{J}/4}$, at most $\frac{1}{2}$ beyond an absolute J. Hence $\mathbb{E}\max_j(S_B^{(j)})^+ \ge \frac{1}{2}u\sqrt{B} = \Omega(\sqrt{B\log J})$. Second, if $\log J > B/8$ and $B \ge 1024$, take $u = \sqrt{B}/32$, so $u\sqrt{B} = B/32$ and $e^{-64u^2} = e^{-B/16}$, and $J \ge e^{B/8}$ makes $J\,e^{-B/16}$ diverge, giving $\mathbb{E}\max_j(S_B^{(j)})^+ = \Omega(B)$, which is $\Omega(\sqrt{B\min\{1+\log J, B\}})$ in that regime. Third, when $\log J < 128$ or $B < 1024$ the quantity $\sqrt{B\min\{1+\log J, B\}}$ is $O(\sqrt{B})$, and a single walk already gives $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \ge \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \ge \frac{1}{2}\sqrt{B/3}$ by Hölder, since $\mathbb{E}S_B^2 = B$ and $\mathbb{E}S_B^4 \le 3B^2$. Combining the three regimes, $\mathbb{E}\max_j(S_B^{(j)})^+ = \Theta(\sqrt{B\min\{1+\log J, B\}})$. This is the route by which maxima of independent random walks fix the minimax rate for prediction with expert advice (Cesa-Bianchi and Lugosi, 2006).

Proof idea. Divide time into blocks of length d and give J actions private states whose losses are perturbed by independent Rademacher signs within each block. Every observation available when an action is chosen comes from an earlier block, and the supplied $m_t = c_{t-d}$ depends only on earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is therefore $T/2$, while the best fixed action in hindsight benefits from the maximum of J independent random walks. Choosing the amplitude to make $E_2 \le \mathcal{E}$ gives the stated scale.

Proof of Theorem 2. Take $h = d$ and $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$. By Lemma 13 every algorithm's expected loss equals $T/2$, since its action is independent of the current block's signs and all states have mean $\frac{1}{2}$ marginally. Its regret is therefore exactly the comparator's fluctuation advantage, $\varepsilon d\,\mathbb{E}\big[\max_{j\le J}\big(\sum_{2\le b\le B}\sigma_b^{(j)}\big)^+\big]$ with $B = \lfloor T/d\rfloor$, over the $B-1$ randomized blocks. Per round $\|c_t - m_t\|_\infty \le 2\varepsilon$, because the supremum norm charges only the largest coordinate and each coordinate moves by at most 2ε at a block boundary; the gap is ε across the first randomized block, where m_t is still the neutral c_1, and lies in $\{0, 2\varepsilon\}$ thereafter. Hence $E_2 \le 4\varepsilon^2 T = \mathcal{E}$ holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 14 and the discussion following it, the expected maximum of the positive parts of J independent Rademacher sums of length $B-1$ is $\Theta(\sqrt{(B-1)\min\{1+\log J, B-1\}})$, which covers both regimes and the case $J = 1$. The hypothesis $d \le T/2$ gives $B \ge 2$ and hence $B-1 \ge B/2$, so replacing $B-1$ by B costs a factor at most $\sqrt{2}$ in each branch of the minimum, including the saturated branch where the minimum is $B-1$ rather than $1+\log J$. Substituting $B = \lfloor T/d\rfloor$ and ε therefore gives $\Theta(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$. Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the $\sqrt{KT}$ term is Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class. $\square$

The other term such constructions produce, the multiplicative $\sqrt{\min\{|\mathcal{S}|, d\}T}$, is not forced here; Appendix K records why the known route to it is unavailable in this model class, and that neither a matching upper bound nor a lower bound ruling it out is established.

Proposition 15. *Suppose $X_t = c_t(A_t)$. On instances whose drift is carried by a single action, $\Lambda_2 = E_2$, so the bound of Bar-On and Mansour (2025) reads $\widetilde{O}(\sqrt{KT + d\,E_2})$ and its drift contribution $\sqrt{d\,E_2}$ matches Theorem 2 up to logarithmic factors.*

In words, single-action drift makes the Euclidean and supremum measures agree, so on that family the drift term of the published upper bound is unimprovable in order. Its $\sqrt{KT}$ term is not matched because the oracle supplies an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an E_2-adaptive upper bound under noisy outcomes.

Proof of Proposition 15. If only one coordinate of $c_t - m_t$ is nonzero then $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$. On the family of Section 4 that coordinate has magnitude at most $2\varepsilon \le 1$, so the truncation at one in Λ_2 is inactive and $\Lambda_2 = E_2$. The bound of Bar-On and Mansour (2025) then reads $\widetilde{O}(\sqrt{KT + dE_2})$, whose $\sqrt{dE_2}$ contribution Theorem 2 matches up to the logarithmic factor at $J = 1$. Its $\sqrt{KT}$ contribution is not matched. $\square$

G The two-action counterexample

Take $K = 2$ with the hybrid regularizer of Zimmert and Seldin (2020), Tsallis parameter $\beta = 0.01$, learning rate $\eta = 0.5$ and residual scale $C = 1$. The iterate solves the stationarity condition $g(x_1) + L_1 = g(x_2) + L_2$ with $x_1 + x_2 = 1$, where

$$g(x) = -\tfrac{1}{2\beta}x^{-1/2} + \tfrac{1}{\eta}\log x$$

and L_a is the cumulative estimate for action a. Suppose the cumulative estimates place action 1 at probability x_1. A single maximally negative centered residual $-C/x_1$ is then delivered to action 1. This is realizable in the model of Section 3, since $m_t(1) = 1$ and $X_t = 0$ give exactly $\check{c}_t(1) - m_t(1) = -C/x_1$ with $C = 1$. Solving the stationarity condition again gives the following.

x_1 before	x_1 after	ratio
9.15×10^{-3}	1.46×10^{-2}	1.6
9.83×10^{-4}	7.39×10^{-3}	7.5
2.19×10^{-4}	9.98×10^{-1}	4561
5.80×10^{-5}	1.00	17226
4.86×10^{-6}	1.00	205641

The transition sits at $4\beta^2 C^2 = 4 \times 10^{-4}$, which is where the Tsallis term $x^{-1/2}$ ceases to dominate the entropy term $\log x$ in g, so the one-step linearization underlying a constant-factor bound fails. The violated statement is the local-stability inequality asserting that $x_{t'}(a)/x_t(a)$ stays bounded by a constant for all t' within d rounds of t. A single round refutes it, so no accumulation argument is needed. This invalidates that lemma and the proof route resting on it, and not every argument that might use the same regularizer. It does not refute the open problem of Section 6, which concerns the regret rate and may be provable by other means. Devices that bound the residual magnitude introduce either a first-order bias dominating the target term or a probability floor narrowing the delay regime.

H Numerical detail

Sixteen studies were run. The six that bear on the claim are reported at length below, each with an observation, an interpretation and a limitation; the other ten are summarised in one table at the end. Studies 1 to 13 draw P from a Dirichlet and 14 to 16 do not. STATE-EXP3 is beaten in study 15, and in four of six cells of study 16 when the baseline alone is grid-tuned. Study 10 is a control for study 6 and reverses its conclusion. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round r's outcome is consumed before round $r + d + 1$, matching Section 3. Reported $\pm$ is one standard error of the paired difference. Throughout this appendix, *sample-path pseudo-regret* means $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$, evaluated on one realization of the learner and environment randomness. Averaging it over seeds estimates the expected pseudo-regret that the theorems bound. We use that name rather than "realized regret" to distinguish it from regret computed from the sampled outcomes X_t, which is not what these studies report.

Study 1. The bound holds and pooling helps. With $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, drift band 0.05 and 100 seeds:

K	state, $m = d+1$	state, $m = 1$	action, $m = 1$	paired gain	state better in
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
20	618.3	493.7	547.5	53.8 ± 5.6	87/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

Observed. Sample-path pseudo-regret for the analyzed variant stayed below $\sqrt{2(d+1)|\mathcal{S}|T \log K}$ in 100 of 100 seeds at every K, the bound running from 2913 to 4228 against realized 455 to 780. The plain variant beat the action-level baseline at every K, by a margin growing from 19 to 169 as K rose tenfold with $|\mathcal{S}|$ fixed, and it beat the analyzed variant throughout. Sweeping the delay at $K = 40$ gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at $d = 10$, 50 and 200.

Interpretation. The margin growing in K at fixed $|\mathcal{S}|$ is the prediction that separates the two estimators, so the improvement is attributable to the state channel the effective dimension attributes to it. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what an additive delay term predicts and a multiplicative one does not, which is the diagnostic evidence behind the open problem of Section 6.

Limitation. This is one synthetic family with a fixed drift band, the plain variant carries no guarantee, and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed below it in the runs performed.

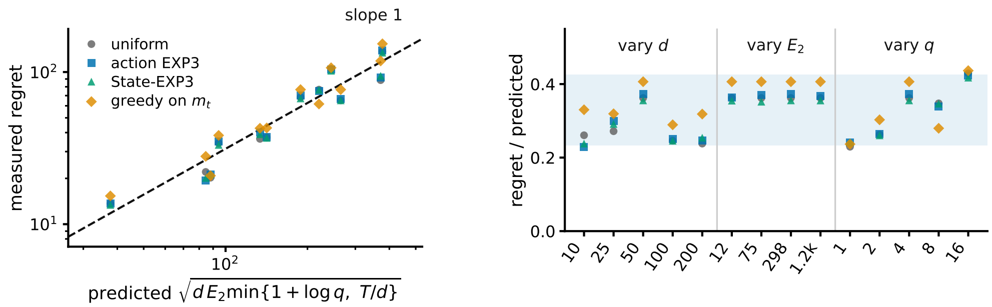

Figure 3: Study 8, the drift channel. **Left,** measured regret against the scale Theorem 2 predicts, over twelve cells in which d varies 20-fold, E_2 100-fold and J 16-fold. The dashed line has slope one. **Right,** the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

Study 6. The effective dimension, not $|\mathcal{S}|$, governs the gain. Holding $|\mathcal{S}| = 8$, $K = 40$, $T = 5000$, $d = 20$ and 24 seeds, and varying only the overlap between the rows of P, the measured mean of v_t moves while $|\mathcal{S}|$ does not:

row overlap	mean v_t	state-pooled	action-level	paired gain
0.00	2.719	392.7	599.3	206.6 ± 20.0
0.25	2.013	361.1	528.1	167.0 ± 21.3
0.50	1.532	306.5	406.5	100.0 ± 10.5
0.75	1.168	199.9	229.2	29.3 ± 3.2
0.95	1.009	49.2	50.8	1.6 ± 0.5

Observed. Both the regret and the advantage over action-level weighting fall monotonically with v_t, from a paired gain of 206.6 at $v_t = 2.719$ to 1.6 at $v_t = 1.009$. Nothing about the raw state count changes across the sweep.

Interpretation. $|\mathcal{S}|$ is constant and cannot explain the pattern, while v_t moves with it, which is the comparison Theorem 5 predicts.

Limitation. The two estimators do not become the same object at $v_t \approx 1$. Rather the rows of P agree, so every action induces the same loss and both regrets collapse, making that endpoint uninformative rather than favorable. One parameter of one map family was varied, so the sweep is consistent with the effective dimension governing the gain rather than establishing it.

Study 8. The drift bound describes the right scaling. The construction of Theorem 2 at $K = 40$, $T = 8000$, $h = d$ and 12 paired seeds, sweeping d from 10 to 200, the amplitude ε from 0.02 to 0.20 so that E_2 runs from 12 to 1193, and the drifting coordinates J from 1 to 16. We run four policies: uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector m_t, the last because Theorem 2 claims to survive it. E_2 is measured on the realized instance rather than predicted, so the normalization uses the quantity the theorem uses.

Observed. The predicted scale $\sqrt{d\,E_2 \min\{1 + \log J, T/d\}}$ ranges from 38 to 377 across the grid and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429, 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and 0.310 for the three that never see m_t.

Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. That greedy play on the exact m_t sits at the top of the band rather than below it is the concrete content of the claim that the bound survives the oracle.

Limitation. Every cell has $T/d > 1 + \log J$, so only the $1 + \log J$ branch of the minimum is exercised and the T/d saturation branch, which needs d comparable to T, is not probed. A lower

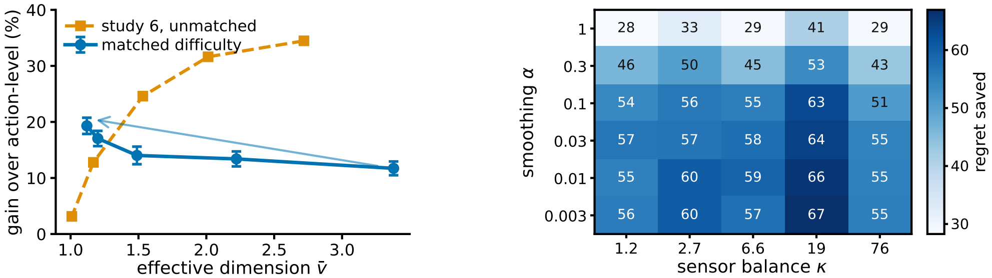

Figure 4: **Left,** study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in Study 6 is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right,** regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell, from the plug-in sensitivity study summarised in the closing table.

bound quantifies over all algorithms and four policies cannot establish that. What these runs check is the scaling of the construction.

Code and environment. The code producing every reported number and figure is at `https://github.com/melikabaghi/state-exp3`. It pins the software environment used for the reported numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each study, lists one command per numbered study, and commits the stored outputs each figure reads so that every figure rebuilds without rerunning an experiment.

The remaining studies in brief. Studies 2–5, 7, 9, 11–13 and 15 are summarised here rather than reported at length, since none of them bears directly on the claim of Section 5. Each was run under the same conventions, and the numbers below are what the runs produced.

Study and finding	What the runs showed
2. An unknown map costs little	The online plug-in stays within 0.5 per cent of known P at every K and beats action-level weighting by 20 to 179 as K rises tenfold; forced exploration never helped, $\gamma = 0$ being best at every K.
3. The plug-in survives hostile maps	On five deliberately hostile maps it still beat action-level weighting, two of them with κ above 100, and forced exploration cost regret in every cell. The Laplace default is not the best smoothing.
4. Safe blending does not recover the gain	No. The rule reproduces action-level regret to the digit, so its guarantee is safe and the algorithm is inert.
5. Counting the denominator is noisier	Counting is the noisier of the two on regret and on bias, and the predicted relative error is vacuous from $K = 40$ through $K = 4000$.
7. The representation has a bias-variance trade	Yes. Regret is lowest at four groups, the number of latent clusters, and stays there under heterogeneity 0.12.
9. The rotating bound tracks the shape, not the constant	Partly. The ratio $R/\sqrt{(d+1)\widehat{v}T\log K}$ spans 35.9 times over the grid, 7.1 times once a near-degenerate column is dropped, so the bound tracks the shape and not the constant.
11. The plug-in is not fragile in the smoothing constant	No cell of thirty loses to action-level weighting, and the gain roughly doubles as α falls from 1 to 0.03.
12. The certificates are far looser than the method	The plug-in stays within 1.6 per cent of known P, while the certificates themselves are far looser than the measured arms.
13. The coarsening cannot be chosen by this rule	No. The rule selects 15 groups at heterogeneity 0 and 16 at 0.12 against a true count of four. Learning g remains open.
15. A best-state rule leads seven of the nine settings	A best-state rule leads seven of the nine settings and STATE-EXP3 two, but where the rule's assumption fails its regret grows linearly and its spread across seeds is four times ours.

Study 10. Overlap drives the gain, not easier tasks. Study 6 sweeps the overlap and finds the advantage falling as $\widehat{v}$ falls, which read at face value contradicts Theorem 5, whose bound improves as $\bar{v}$ falls. That sweep is confounded, since raising the overlap also flattens $c_t = P\theta_t$ across actions,

collapsing the comparator gap so that both regrets vanish together. This study holds $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$ and the difficulty fixed, and varies only $\widehat{\bar{v}}$. Difficulty is the uniform-play regret $\sum_t \mathrm{mean}_a c_t(a) - \min_a \sum_t c_t(a)$, a property of the instance with no algorithm in it, and the amplitude of θ is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and θ is aligned with it, which is what lets the gap survive as the rows agree.

Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at $\widehat{\bar{v}} = 3.380$ to 19.3 per cent at $\widehat{\bar{v}} = 1.120$, with absolute gains 85.5 ± 8.9 to 106.2 ± 7.9 and a correlation of -0.929 between $\widehat{\bar{v}}$ and the gain. Study 6's unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range (Figure 4).

Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the overlap dependence suggested by Theorem 5, although this experiment uses the practical $m = 1$ variant rather than the theorem's $m = d + 1$ algorithm. Overlap is what pooling exploits, and study 6 measures overlap confounded with an instance becoming trivial. This also explains study 9's degenerate column without appealing to it.

Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller gap should if anything shrink the advantage, the measured rise is conservative in that direction. The matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the difficulty costs.

Study 10b. The matched-difficulty trend survives a second configuration. Study 10 is a single design point, so the same procedure was re-run at $K = 20$, $|\mathcal{S}| = 12$, $T = 8000$, $d = 20$ and 20 paired seeds, with the common target lowered to 500 so that it is reachable at every overlap. Nothing else changes.

overlap	$\widehat{\bar{v}}$	R_{unif}	state	action	paired gain
0.00	3.440	500.0	364.7	407.2	42.5 ± 8.2
0.35	2.225	500.0	346.8	401.3	54.6 ± 8.3
0.65	1.518	500.0	309.3	372.4	63.2 ± 6.4
0.85	1.240	493.6	271.8	346.0	74.3 ± 7.2
0.95	1.191	490.1	256.9	327.6	70.7 ± 7.8

Observed. The relative gain rises from 10.4 to 21.6 per cent as $\widehat{\bar{v}}$ falls from 3.440 to 1.191, and the correlation between $\widehat{\bar{v}}$ and the absolute gain is -0.972 against study 10's -0.929. Realized difficulty spans 490.1 to 500.0, a 1.020 times spread, so the matching is tighter here than in study 10.

Interpretation. The direction study 10 reports is not an artifact of its single configuration. Two independent design points, differing in K, $|\mathcal{S}|$, T and d, give the same sign and a comparable magnitude.

Limitation. Both configurations use the same construction and the same matching routine, so this tests configuration sensitivity and not construction sensitivity. Both run the practical $m = 1$ variant. An earlier attempt at this replication silently failed because `match_scale` binds its target as a default argument, so the bisection saturated and the difficulty was never matched; the run reported here passes the target explicitly.

Study 14. Many actions resolved by few states is a natural regime. Every study so far draws P from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are $K = 200$ catalogue items, states are $|\mathcal{S}| = 6$ ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

Observed. The effective dimension is $\widehat{\bar{v}} = 1.97$ against $K = 200$, a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 5). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at $d = 10$, 50 and 200, the largest margins anywhere in this appendix. The plug-in matches known P at every delay, the paired

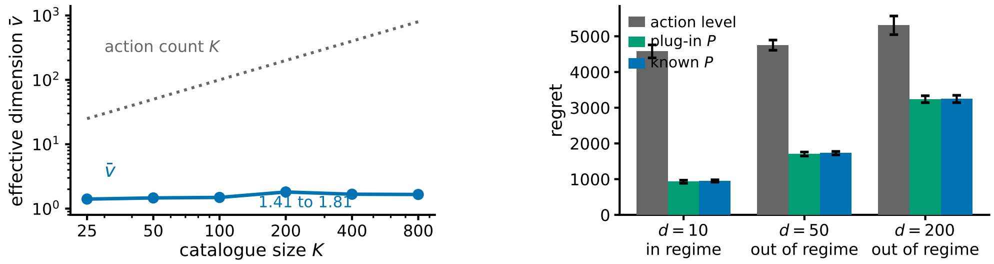

Figure 5: Study 14, a recommendation funnel. **Left,** the effective dimension stays near two while the catalogue grows thirty-twofold, so the gap between K and $\widehat{v}$ comes from the catalogue having categories, not from a chosen K. **Right,** the three arms at $T = 20000$ and 8 paired seeds, labelled by whether the regime condition of Section 4 holds. Bars are one standard error. No real data is used anywhere in this study.

difference never reaching two standard errors. The regime condition $\bar{v}(d+1)\log K \leq K + d$ holds at $d = 10$ and fails at $d = 50$ and $d = 200$.

Interpretation. The separation between K and $\widehat{v}$ arrives from catalogue structure rather than from a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet families cannot make. That the condition Theorem 5 needs fails at the delays a funnel has, while the practical variant still saves most of the regret, is the same message as study 1 in the setting that motivates the paper. It is evidence for the open problem of Section 6 rather than for the theorem.

Limitation. *No real data is used.* There is no interaction log in this environment, nothing here is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The structure is imposed by hand from qualitative properties, so it shows that the regime is describable and self-consistent, not that any deployed system occupies it. The arms run $m = 1$, which carries no guarantee, and the environment has no adversarial component, so it exercises the pooling channel and not the drift channel.

Study 16. Pooling still helps against the strongest published algorithm. Every comparison so far is against action-level weighting, EXP3-IX, naive sharing, or the best-state rule. None is the strongest published method for delayed feedback: that is the hybrid Tsallis-plus-entropy FTRL of Zimmert and Seldin (2020), whose delay term is additive rather than multiplicative. We run it on both headline families, with the same environments, seeds and delay convention, under three tunings.

family, cell	STATE-EXP3	Zimmert and Seldin	same, grid-best	STATE-EXP3, grid-best
funnel, $d = 10$	949.0	2959.4	2413.8	27.0
funnel, $d = 50$	1728.1	3557.0	2556.4	95.9
funnel, $d = 200$	3244.8	4734.7	2807.7	199.7
overlap 0.00	661.5	731.8	610.4	202.4
overlap 0.65	584.1	673.1	544.0	51.2
overlap 0.95	461.4	540.2	419.4	34.3

Observed. With STATE-EXP3 at the single tuning used everywhere else here and Zimmert and Seldin (2020) at the best of a six-point grid around its own value, STATE-EXP3 wins every cell, cutting regret by 31.5 to 67.9 per cent on the funnel and 9.6 to 14.6 per cent on the matched sweep. With both at their grid-best, over seven multipliers for Zimmert and Seldin (2020) and six for STATE-EXP3, STATE-EXP3 wins every cell again and by more, 66.8 to 98.9 per cent. With Zimmert and Seldin (2020) at its grid-best and STATE-EXP3 left at its paper tuning, the third column against the first, Zimmert and Seldin (2020) wins the funnel at $d = 200$ and all three matched cells, four of six.

Interpretation. No column is a symmetric protocol. The two grids are finite and unequal, and STATE-EXP3's grid-best sits at its top multiplier in every cell, so its own minimum may lie outside the range searched; the truncation runs against STATE-EXP3. Where it is not held to the worse tuning of the two, the state channel helps against the strongest published method, which is expected, since that method is action-level by construction and cannot read the state at all. Grid tuning helps STATE-EXP3 far more than it helps Zimmert and Seldin (2020), which is consistent with pooling cutting estimator variance and lower variance admitting a larger learning rate.

Limitation. The third column against the first is a real loss and is reported as one. These are our own synthetic families, and a method built for adversarial worst cases is not shown at its best on benign instances. Every grid-best multiplier is chosen by reading these benchmarks, so none of the last three columns is a procedure a practitioner could run; only the first is. Eight paired seeds on the funnel and ten on the matched sweep.

Taken together the sixteen studies cover both channels. Studies 1, 6, 7, 9, 10, 14 and 15 probe the state channel and are consistent with the effective dimension, rather than $|\mathcal{S}|$ or K, governing what pooling buys; study 10 had to control for task difficulty before study 6's sweep pointed that way, and study 9 shows where the interleaved bound stops describing the dependence. Studies 2 to 5, 11 and 12 estimate the action-to-state matrix and locate the difficulty in certifying the estimate rather than in forming it. Study 13 shows the grouping bound is too conservative to serve as its own selection rule. Study 14 is the only one whose matrix is not drawn from a Dirichlet and reports the largest margins here, on a structure imposed by hand rather than measured. Study 15 finds STATE-EXP3 beaten in seven of nine settings by a heuristic that assumes the best action reaches the single best state, which fails with linear regret where that assumption does not hold. Study 16 places it against the strongest published method: ahead in every cell when both are tuned alike, behind in four of six when that method alone is grid-tuned.

I A route that does not work

Theorem 1 is proved in Appendix J by bounding a state-marginal ratio. The natural first attempt is different and fails, which is worth recording so the detour is not repeated.

Run STATE-EXP3 with $m = 1$ and let $\tilde{x}_r$ be the iterate that has also absorbed the outstanding estimates $G_r = \sum_{s=r-d}^{r-1} \hat{c}_s \geq 0$. The delay term $\mathbb{E}[\langle x_r - \tilde{x}_r, \hat{c}_r \rangle]$ is at most $\eta \sum_{s=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_s \rangle] \leq \eta d$ for any unbiased estimator, which is already the additive form. The quadratic term is where it stops: the telescoping scores $\sum_a x_{r+d}(a)\hat{c}_r(a)^2$ while Proposition 4 bounds the sum against x_r, and the substitution costs $|\mathcal{S}|/Z_r$ with $Z_r = \mathbb{E}_{a\sim x_r}[e^{-\eta G_r(a)}]$, hence an exponential moment of $\langle x_r, G_r \rangle$.

That moment has no bound uniform over state distributions. Here $\langle x_r, \hat{c}_s \rangle = q_r(S_s)X_s/q_s(S_s)$, governed by the reciprocal probability of the state actually observed, whose mean is $|\mathcal{S}|$ but whose exponential moment is not uniformly finite: already at $|\mathcal{S}| = 2$ with $q = (\varepsilon, 1-\varepsilon)$, $\mathbb{E}[e^{\eta/q(S)}] \geq \varepsilon e^{\eta/\varepsilon} \to \infty$ while the mean stays at 2. Flooring q restores the moment and costs the product back, since $\eta\langle x_r, G_r \rangle \leq 1$ then needs $\gamma \asymp \eta d\kappa|\mathcal{S}|$ and the mixing cost γT reproduces $\widetilde{O}(\sqrt{d\kappa|\mathcal{S}|T\log K})$. Empirically the obstruction never bites, $1/Z_r$ staying below 1.33 at the tuned η over $d \in \{10, 50, 200\}$, which is what suggested it was an artifact of the device rather than of the problem. That the ratio really can blow up, rather than merely resisting a bound, is shown in Appendix G, where a two-action instance under a hybrid regulariser moves an iterate by a factor of 2×10^5 in one step.

J Proof of Theorem 1: additive delay without interleaving

Interleaving cannot give an additive bound. Copy i, which absorbs a share V_i of the budget $V = \sum_i V_i$, has regret $\sqrt{2V_i \log K}$, and Cauchy–Schwarz over the m copies gives $\sqrt{2mV\log K}$, tight when the V_i agree, so the $\sqrt{m}$ in Theorem 5 is not an artifact of the analysis. An additive bound has to come from a single copy.

Throughout, $L_t = \sum_{r\leq t-d-1} \hat{c}_r$ and $x_t \propto e^{-\eta L_t}$, so $L_{t+1} - L_t = \hat{c}_{t-d}$, with $\hat{c}_r = 0$ for $r < 1$. The filtration $\mathcal{F}_t$ is the environment's, including outcomes not yet delivered, as in Proposition 4; under it x_t is $\mathcal{F}_{t-d-1}$-measurable, so x_{r+d} is $\mathcal{F}_{r-1}$-measurable.

The drift lemma. Appendix I priced the change of measure from x_r to x_{r+d} through $1/Z_r$. That is avoidable. The pooled estimate satisfies

$$\langle x_t, \hat{c}_t \rangle = \sum_a x_t(a) \frac{P(S_t \mid a) X_t}{q_t(S_t)} = X_t \le 1$$

identically, on every path, because the denominator is the state marginal of the same distribution that supplies the numerator.

Lemma 16. *If $\eta \le 1/(e(d+1))$ then $x_{t+1}(a) \le (1+1/d)\, x_t(a)$ for every t and every a, surely, and hence $x_{t+d}(a) \le e\, x_t(a)$.*

Proof. Strong induction on t. For $t \le d$ nothing is delivered and $x_{t+1} = x_t$. For $t > d$, assume the claim for every $s < t$ and compose it over the d steps from $t-d$ to t, giving $x_t(a) \le (1+1/d)^d x_{t-d}(a) \le e\, x_{t-d}(a)$. Contracting with the fixed nonnegative vector $\hat{c}_{t-d}$ and using the identity above, $\langle x_t, \hat{c}_{t-d} \rangle \le e \langle x_{t-d}, \hat{c}_{t-d} \rangle = e X_{t-d} \le e$. With $Z_t = \sum_b x_t(b) e^{-\eta \hat{c}_{t-d}(b)}$ and $e^{-z} \ge 1 - z$, this gives $Z_t \ge 1 - \eta \langle x_t, \hat{c}_{t-d} \rangle \ge 1 - \eta e \ge d/(d+1)$, so $x_{t+1}(a) = x_t(a) e^{-\eta \hat{c}_{t-d}(a)}/Z_t \le x_t(a)/Z_t \le (1+1/d) x_t(a)$. □

What the induction bounds is $\langle x_t, \hat{c}_{t-d} \rangle = X_{t-d}\, q_t(S_{t-d})/q_{t-d}(S_{t-d})$, the state-marginal ratio Appendix I could not reach. Individual $\hat{c}_r(a)$ remain unbounded; only their x_t-weighted mass is controlled. Cesa-Bianchi et al. (2019) prove the action-level version under $\eta \le 1/(Ke(d+1))$ and Thune et al. (2019) remove the K, using that their estimate is carried by the single action played, so the weight cancels into a same-action ratio. A pooled estimate is spread over every action and does not collapse that way; the identity above replaces the cancellation.

Centring. Let $\zeta_t = \hat{c}_t - X_t \mathbf{1}$, with $\zeta_r = 0$ for $r < 1$. Exponential weights is unchanged by a shift that is constant across actions, so x_t is the same iterate and this is a change of analysis, not of algorithm (Eldowa et al., 2024). Keeping $\gamma(s) = \mathbb{E}[X_t^2 \mid S_t = s]$ inside the state sum,

$$\mathbb{E}\big[\langle x_t, \zeta_t^2 \rangle \mid \mathcal{F}_{t-1}\big] = \sum_s \gamma(s) \Big[\frac{\sum_a x_t(a) P(s \mid a)^2}{q_t(s)} - q_t(s) \Big] \le v_t - 1,$$

each bracket being nonnegative by Jensen and $\gamma(s) \le 1$, the final step being Remark 3. Bounding $\langle x_t, \hat{c}_t^2 \rangle$ by v_t first and then subtracting does not work: it leaves $v_t - \mathbb{E}[X_t^2]$.

Proof of Theorem 1. Since $\zeta_r(a) \ge -X_r \ge -1$ we have $\eta \zeta_r(a) \ge -\eta$, and Lagrange's remainder gives $e^{-z} \le 1 - z + (e^\eta/2) z^2$ for $z \ge -\eta$. With $\log(1+y) \le y$, the potential $W_t = \sum_a \exp(-\eta \sum_{r \le t-d-1} \zeta_r(a))$ obeys $\log(W_{t+1}/W_t) \le -\eta \langle x_t, \zeta_{t-d} \rangle + (\eta^2 e^\eta/2) \langle x_t, \zeta_{t-d}^2 \rangle$. Telescoping from $W_1 = K$ and using $\log W_{T+1} \ge -\eta \sum_{r \le T-d} \zeta_r(a^*)$ for the comparator a^*,

$$\sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r \rangle \le \sum_{r=1}^{T-d} \zeta_r(a^*) + \frac{\log K}{\eta} + \frac{\eta e^\eta}{2} \sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r^2 \rangle.$$

The shift cancels between the two sides. As x_r is $\mathcal{F}_{r-d-1}$-measurable, $\mathbb{E}[\langle x_r, \hat{c}_r \rangle] = \mathbb{E}[\langle x_r, c_r \rangle]$ and $\mathbb{E}[\hat{c}_r(a^*)] = c_r(a^*)$, and the d rounds undelivered by $T+1$ contribute at most d, so

$$\mathbb{E}[R_T] \le \frac{\log K}{\eta} + \frac{\eta e^\eta}{2} \sum_r \mathbb{E}\big[\langle x_{r+d}, \zeta_r^2 \rangle\big] + \sum_r \mathbb{E}\big[\langle x_r - x_{r+d}, \hat{c}_r \rangle\big] + d.$$

Lemma 16 gives $x_{r+d} \le e\, x_r$ surely, and x_{r+d} is $\mathcal{F}_{r-1}$-measurable, so the second sum is at most $e \sum_r \mathbb{E}[v_r - 1] \le e V^-$.

For the third, write $x_{r+d}(a) = x_r(a) e^{-\eta G_r(a)}/Z'$ with $G_r = \sum_{u=r-d}^{r-1} \hat{c}_u \ge 0$ and normaliser $Z' \le 1$, distinct from the one-step Z_t of the Lemma. Then $x_r(a) - x_{r+d}(a) \le x_r(a)(1 - e^{-\eta G_r(a)}) \le \eta x_r(a) G_r(a)$ for every a, trivially where the left side is negative. Since $\hat{c}_r \ge 0$ and x_r, G_r are $\mathcal{F}_{r-1}$-measurable, the conditional expectation is at most $\eta \langle x_r, G_r \rangle$, and $\mathbb{E}[\langle x_r, G_r \rangle] = \sum_{u=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_u \rangle] \le d$: for each such u, x_r is $\mathcal{F}_{r-d-1}$-measurable and $r - d - 1 \le u - 1$, so it leaves the conditional expectation and $\mathbb{E}[\langle x_r, \hat{c}_u \rangle] = \langle x_r, c_u \rangle \le 1$. This is measurability, not independence, and at $u = r - d$ the two σ-fields coincide, so the step rests on the timing convention of Section 3. The sum is at most ηdT.

Finally $\eta \le 1/(2e)$ gives $e^{1+\eta} < 3.3$, so $\mathbb{E}[R_T] \le \log K/\eta + \frac{\eta}{2}(3.3V^- + 2dT) + d$. For $f(\eta) = A/\eta + B\eta$ restricted to $\eta \le \eta_{\max}$ one has $\min f \le 2\sqrt{AB} + A/\eta_{\max}$, in the interior and in the binding case alike; with $A = \log K$, $B = (3.3V^- + 2dT)/2$ and $\eta_{\max} = 1/(e(d+1))$ this is the stated bound. $\square$

The cap on η binds only when $d \gtrsim T/(e^2 \log K)$, and not at the settings of Appendix H: the funnel study's own rates are 0.0058, 0.0031 and 0.0016 at $d = 10, 50, 200$ against caps 0.0334, 0.0072 and 0.0018, so Theorem 1 covers those runs as they were executed. Over three seeds and 20000 rounds the induction has room to spare, $\max_a x_{t+1}(a)/x_t(a)$ reaching 1.006, 1.003 and 1.002 against the permitted 1.100, 1.020 and 1.005.

The two bounds are numerically close on the funnel, where $\bar{v} \approx 2$: the additive form carries $3.3(\bar{v} - 1) + 2d$ and the interleaved one $(d+1)\bar{v}$, and these cross near $\bar{v} = 2$. The additive form gains as $\bar{v}$ rises toward $|\mathcal{S}|$, and it describes the algorithm that is actually run, which is the reason to prefer it.

K Stationary sensors

Observation 17. The multiplicative term $\sqrt{\min\{|\mathcal{S}|, d\}T}$ is not forced by the known construction, and is absent at one endpoint of the present model class.

The $\Omega(\sqrt{\min\{|\mathcal{S}|, d\}T})$ construction of Esposito et al. (2023, Thm. 5.2) re-splits states across actions adversarially per block, so its action-to-state map is time-varying and lies outside the model class of Section 3. At $E_2 = 0$ with stationary P the problem is a stochastic delayed bandit, where delay enters additively (Joulani et al., 2013). One route to it is therefore unavailable and the term is absent at one endpoint, but neither a matching upper bound for all $E_2 > 0$ nor a lower bound ruling it out is established here.